8.3 Spatial Health Inequality in Latin American Cities: Magnitude, Drivers, and Policy Alternatives

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8.3.1 Abstract

Spatial health inequities are pervasive across and within Latin American cities. This article explores the magnitude of spatial inequalities in health outcomes, their drivers, and policies that can be deployed to address them. Life expectancy, the burden of noncommunicable diseases, and, more recently, the impact of COVID-19, all exhibit large geographic disparities. These variations are correlated with individual-level socioeconomic factors, such as low schooling levels and high poverty rates. Nevertheless, spatial health inequalities are not only a reflection of the residential clustering of vulnerable individuals. Local environments themselves can improve or further deteriorate their communities' public health outcomes. These include weather conditions, public utilities, recreational infrastructure, and externalities—notably air and noise pollution. The current evidence suggests that existing policy tools available to local and national governments can be effective in creating healthier built environments, reducing and preventing negative externalities, and promoting healthier lifestyles in cities and neighborhoods.

8.3.2 Introduction

There are large disparities in health outcomes across individuals in Latin America and the Caribbean. These are closely related to socioeconomic conditions. Individuals' education and income, for example, are tied to their ability to access health care services and make healthier lifestyle decisions.

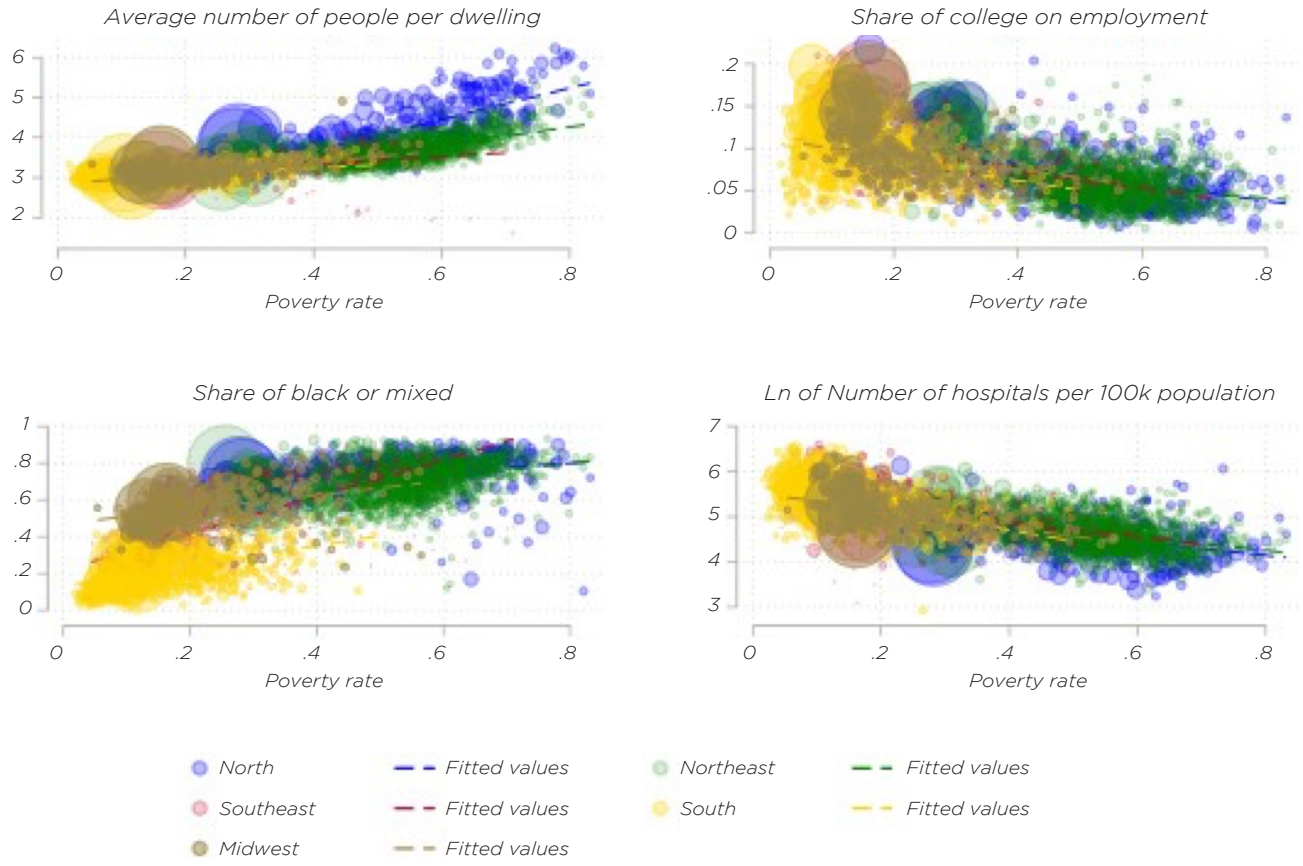
This, in turn, translates into diverging health care outcomes. For example, there is a higher probability of a person having at least two health risk factors (i.e., high cholesterol, hypertension) when that individual's education level is below secondary, relative to those with at least secondary education (Berlinski et al., 2020).

An important feature of these inequalities is that they have a clear spatial dimension. Health outcomes vary, often significantly, across cities and across neighborhoods of the same city. Part of this geographic variation is explained by the fact that people with similar socioeconomic characteristics tend to spatially cluster. This is illustrated in Figure 16, which shows some of the correlates of local poverty rates across Brazilian cities. Higher poverty rates tend to be associated with higher residential overcrowding (number of people per dwelling), lower shares of college in employment, and larger shares of ethnic minorities in the population, which Giuffrida (2007) has shown to be a good predictor of spatial disparities in health outcomes.

Please address related correspondence to juancha@iadb.org. *Nicolás Herrera L., Haydée Svab, and Julio Trecenti provided outstanding research assistance. The opinions expressed in this publication are those of the authors and do not necessarily reflect the views of the Inter-American Development Bank, its Board of Directors, or the countries they represent.*

Figure 16 **Correlates of poverty rate across cities in Brazil**

Note: Authors' calculations using data of 3,762 Brazilian cities with population greater or equal to 5,000. 240 cities from North, 1,454 from Northeast, 1,042 from Southeast, 616 from South and, 295 from Midwest. Cities are weighted by population. Data from Chauvin (2021a).



Although the composition of the population of urban areas explains part of the observed heterogeneity in health outcomes across space, it does not fully account for all the observed variation. Spatial segregation itself can create and perpetuate health disparities. Low-income areas of cities frequently have limited access to key infrastructure such as drinking water, sewerage, and public transit. They also have reduced access to health services, as illustrated in Figure 16 (bottom right), which shows that poorer Brazilian cities tend to also have fewer hospitals per 100 thousand people. This exposes their population to unhealthy environments, leading to worse health outcomes relative to economically better-off areas. Spatial differences in the access to health care services is not the only problem that low-income areas face; the quality of it is also different. Within cities, spatial health inequality goes hand in hand with spatial socioeconomic inequality. Figure 17 provides an illustration showing some of the correlates of bad health quality, defined as the percentage of individuals that considered their health service as "bad" or "very bad" across sub-metropolitan geographic units in Bogotá, Colombia. A poor health service is a feature of areas of the city where there are also high levels of labor informality and high illiteracy rates—both variables associated with low income levels—as well as high residential overcrowding. On the other hand, even after controlling for individual characteristics such as education and income, and the way they affect self-selection into neighborhoods, there is still a significant relationship between health variables and the characteristics of the local environment. For example, using data from Italy, Bilger and Carrieri (2013) point out that people in neighborhoods with low environmental quality are more likely to suffer from chronic diseases. In Latin American cities, security perception and residing in a walkable neighborhood are related with high levels of physical and mental health, while street noise is negatively related with these outcomes (Gomez et al., 2020).

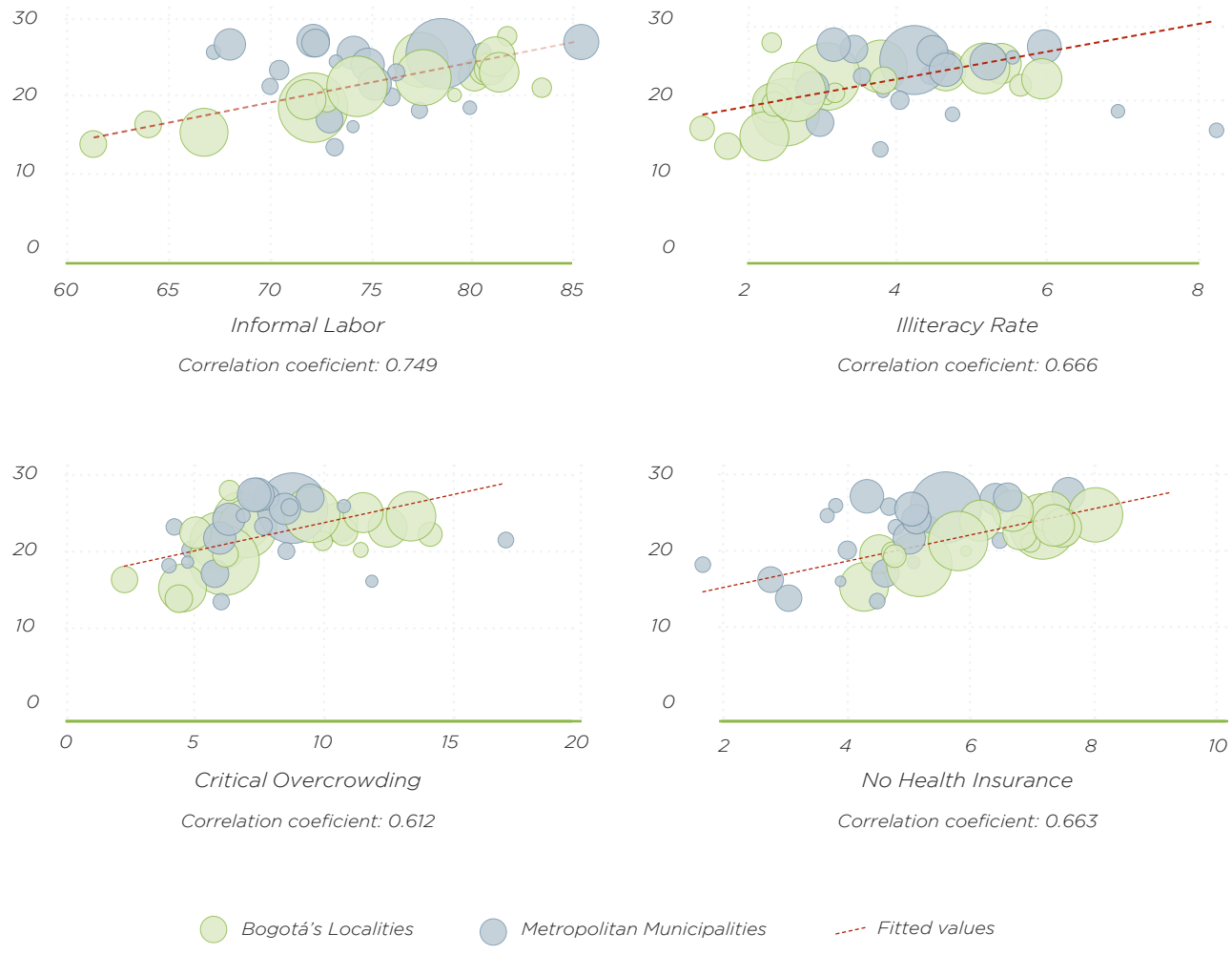
This article presents an overview of the existing evidence on three interrelated questions: How large are spatial inequalities in health outcomes? What are their key drivers? And what can public policy do about them?

Section 2 explores the magnitude of the variations in health outcomes across space (i.e., across cities and neighborhoods in multiple Latin American countries). Section 3 discusses the factors that could explain such urban differences. Section 4 reviews some of the most common implemented policies for addressing these disparities. Last, Section 5 concludes the discussion.

Figure 17

Bad health quality perception and socioeconomic characteristics across Bogotá's Metropolitan Area

Note: Bogotá's metropolitan area includes the 20 localities of Bogotá and 20 surrounding municipalities. Localities include 1922 neighborhoods. Localities and municipalities are weighted by population. Bad health quality is defined as the percentage of individuals that considered their health insurance service as "bad" or "very bad." Critical overcrowding is measured as those households with more than three people per room. People working in enterprises with up to five workers are considered part of the informal labor sector. For other details see DANE (2009, 1). Correlations are computed using data from the Multipropósito Survey (SDP, DANE, and Gobernación de Cundinamarca, 2017) and the 2018 National Census (DANE, 2018).



8.3.3 How large are spatial health disparities?

8.3.3.4 Health disparities across cities

Between 2000 and 2017, Latin America and the Caribbean experienced improvements across a broad range of health outcomes. Life expectancy increased by around four years, while under five-age mortality fell by 46 percent, and infant mortality by 35 percent (OECD/The World Bank, 2020).

Despite this progress, health inequalities persist across cities in the region. For instance, the burden of cardiovascular diseases such as hypertension (HTN) could be alarming for cities such as Buenos Aires (29 percent) but substantially lower in others, like Quito (9 percent) (Hernández-Hernández et al., 2010). Although this may reflect differences in culture and diets between these two countries, national-level data suggest that Argentina's hypertension—32.34 percent—(Díaz and Ferrante, 2015) is not drastically different from Ecuador's—37.2 percent—(Ministerio de Salud Pública, 2019). A closer look suggests that within the same country, there are also remarkable differences across cities. For example, in Argentina the HTN prevalence in La Plata is 32.7 percent but it could reach up to 39.8 percent in General Bengrano (Carbajal, 2013).

In a similar way, overweight in the population ages 0–19 (according to the definition of the International Obesity Task Force, IOTF) is roughly 3.5 times as high in Tijuana, Mexico that in Ouro Preto, Brazil (Rivera et al., 2014). While there are large disparities among cities of different countries, most of the observed spatial inequality in health outcomes appears to be driven by differences across cities of the same country. In a sample of 286 Latin American cities, for example, 57 percent of the variation in infant mortality rates is explained by within-country variation (Ortigoza et al., 2020).

8.3.3.4.1 Disparities in life expectancy

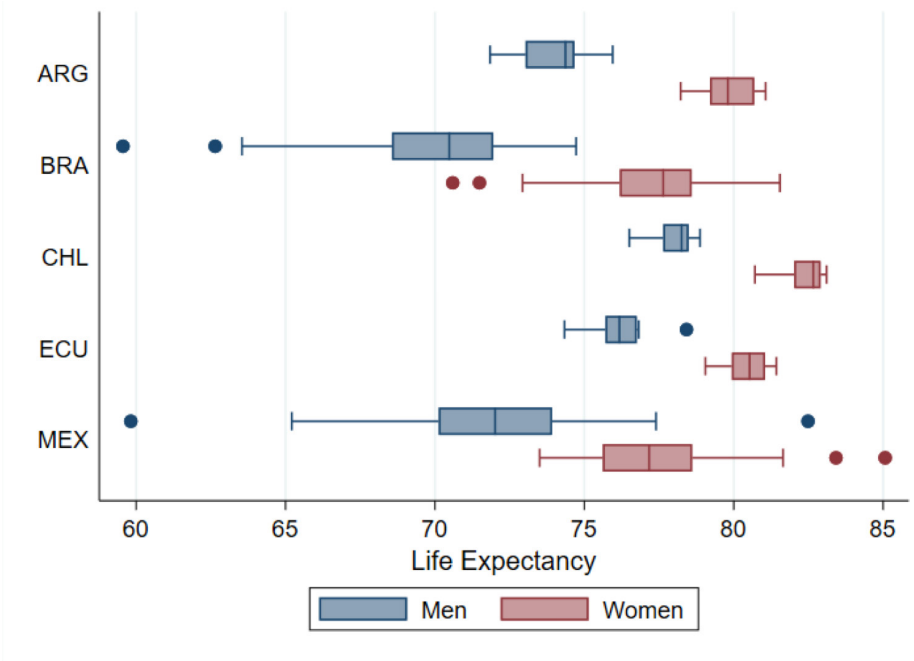
Spatial disparities in morbidity translate into spatial differences in life expectancy. Males have a six-year lower life expectancy in Mexico City than in Santiago, Chile. For females, the gap is of almost eight years (Bilal et al., 2019). In a recent study including urban areas from multiple countries in the region, Bilal et al. (2021a) found that the gap between the cities with the highest life expectancy (David City, in Panama) and the lowest life expectancy in their sample (Juliaca, in Peru), is around 8.3 years.

Figure 18 illustrates these differences by showing the cross-city distributions of gender- specific life expectancy in five countries in the region, using data from 188 cities with a population of at least 200,000. In all countries, females have a higher life expectancy than males, with an average gap of around seven years. The difference between the city with the highest and the lowest life expectancy within each country can be nearly three years for those countries with a more homogeneous life expectancy distribution (Argentina, Chile, and Ecuador), and even higher for other countries, where geographic inequalities tend to be more pronounced. For Mexican men, this difference is almost 23 years. Such disparity is 1.3 times the gap between Canada and Haiti, the top and bottom life-expectancy countries in the region (United Nations and Affairs, 2019). In Brazil, the same difference is around 13 years for both men and women.

Geographic differences in life expectancy are partly explained by the fact that local populations experience disparate health risks and morbidities across cities. For example, the percentage of deaths caused by communicable, maternal, neonatal, and nutritional (CMNN) diseases in Latin American cities can range from 6 to 55 percent. And even more pronounced differences can be found in the share of deaths associated with noncommunicable diseases, which can range between 28 and 71 percent (Bilal et al., 2021b).

Figure 18 ***Differences in life expectancy by gender in five Latin American countries***

Note: Life expectancy is computed for 117 Latin American cities with population greater than or equal to 200,000 (22 from Argentina, 81 from Brazil, 11 from Ecuador, and 63 from Mexico). Cities are weighted by population. The definition of city is comparable across countries. The estimation of life expectancy is done under standard life tables as defined by WHO (2014). Different to Bilal et al., 2021b, age-mortality rates are not discounted by mortality patterns or potential errors in the registration. Mortality and population data come from the ministries of health and national agencies of statistics: INDEC and MSAL (Argentina), IBGE (Brazil), INEC (Ecuador), and INEGI (Mexico).

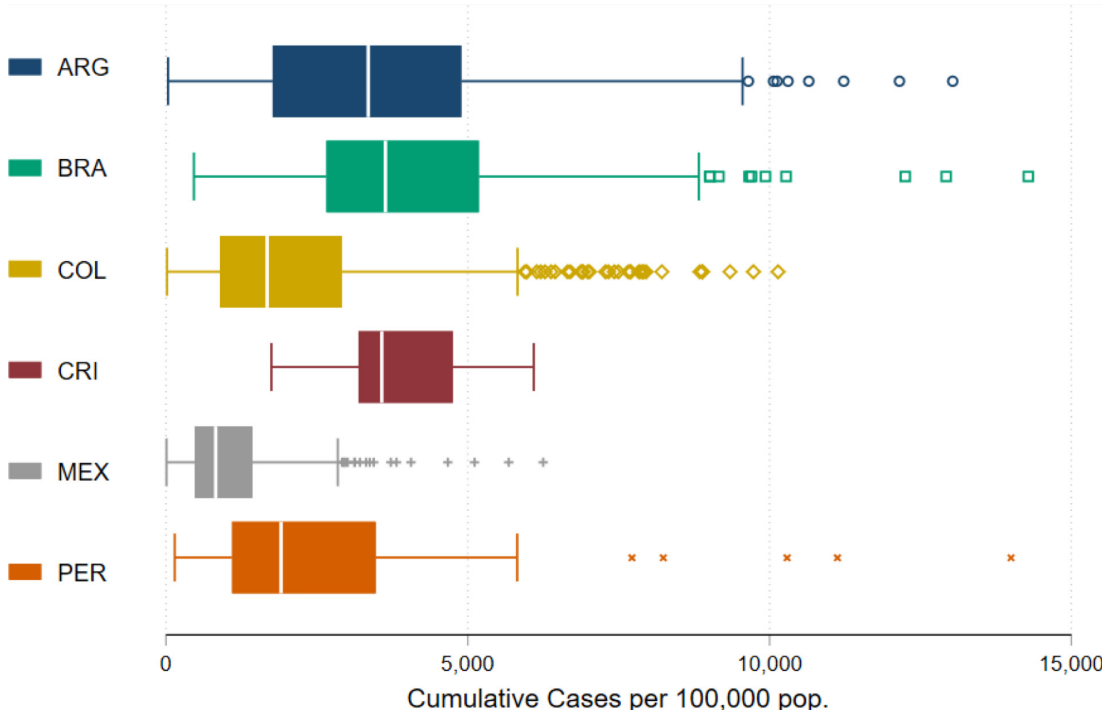


8.3.3.4.2 Disparities in the impact of COVID-19

Cross-city differences in exposure to health risks also became starkly apparent as the COVID-19 pandemic unfolded. The virus first arrived in the continent at the end of February 2020. Brazil was the first country to confirm an infection and would remain as one of the most affected countries in the world in the subsequent months. From the very beginning of the pandemic, the impact of the disease exhibited large variation across cities of the same country. **Figure 19** shows the cross-city distribution of the number of confirmed cases per 100 thousand people in six countries from the region as of January 2021. Part of this variation reflects differences in testing capabilities, but even after addressing these concerns, it is clear that during the first year of COVID-19, the city you lived in mattered for how exposed you were to the virus (Chauvin, 2021b). For Brazilian cities above the 75th percentile of confirmed cases per capita, the incidence was more than four times the number of confirmed cases of the cities at the bottom 25 percent. In Peru and Colombia, half of the cities experienced more than 1,700 cases per 100 thousand inhabitants, which was about two times the Mexican median incidence. In Argentina, the most affected city had 220 percent more cases per capita than the cities at the third quartile.

Figure 19 **COVID-19 confirmed cases per 100,000 people across six Latin American countries**

Note: Cumulative cases per capita are restricted to 1,904 Latin American cities with population greater or equal to 10,000. There are 240 observations from Argentina, 388 from Brazil, 615 from Colombia, 17 from Costa Rica, 463 from Mexico, and 182 from Perú. COVID-19 data are updated until the latest available date per city, as revised on February 10, 2021. The definition of city is not necessarily comparable across countries. For each country the respective unit of analysis is: municipalities for Brazil, Colombia, and Mexico; departments for Argentina; cantons in Costa Rica; and districts in Perú. Data comes from Chauvin (2021a).

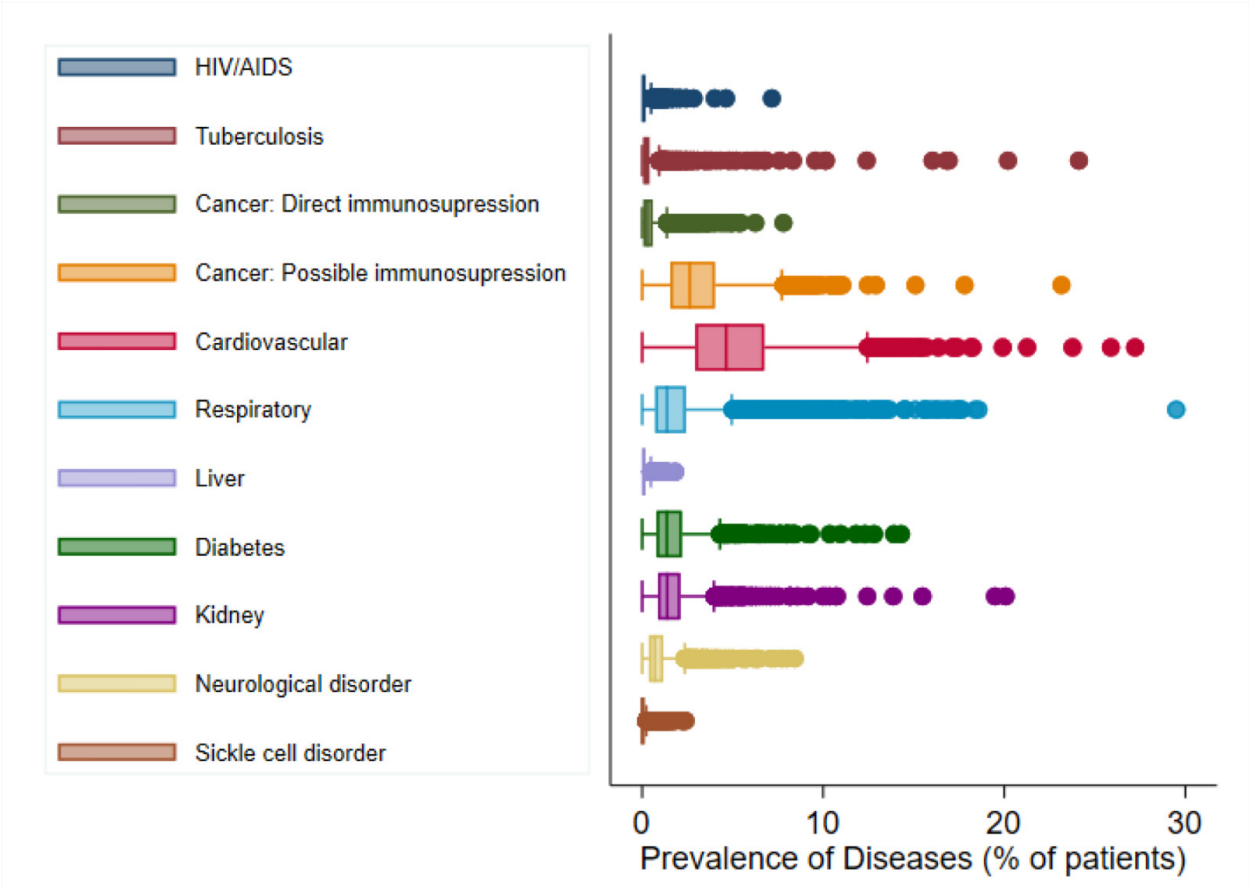


Spatial disparities in health outcomes that existed before the pandemic likely played a role in shaping the unequal geographic incidence of the disease. **Figure 20** depicts, for 3,761 Brazilian cities with population of at least five thousand, the prevalence of pre-existing conditions associated with severe COVID-19 complications among patients in public and private hospitals in 2019. In general, the most common diseases (cardiovascular, respiratory, kidney, tuberculosis, and diabetes) tend to exhibit large geographic variation. For cardiovascular preconditions, cities at the 75 percentile have an incidence that is about 3.5 times those at the 25 percentile. Some cities display a prevalence as high as 28 percent, that is, six times the median. These differences are even more striking for respiratory diseases, as the city with the highest incidence of preconditions (29 percent) has 21 times the median prevalence. Cancer with possible immunosuppression, diabetes, and kidney diseases have a prevalence at the 75 percentile that is around 2.5 times that of cities at the 25 percentile.

In contrast, low-prevalence diseases, such as liver-related or sickle cell disorders, tend to have a more homogeneous distribution across urban areas.

Figure 20 ***Pre-existing conditions across cities in Brazil***

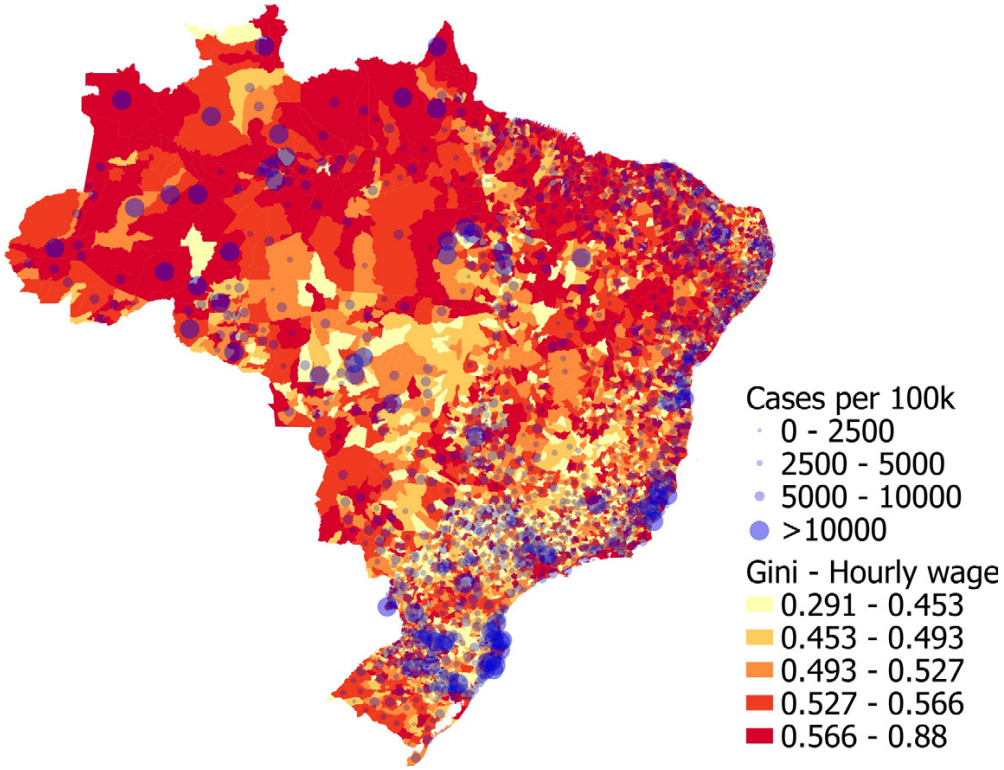
Note: Authors calculations with data of 3,761 Brazilian cities with population greater or equal to 5,000. Data comes from Chauvin (2021b). Prevalence of diseases per patient is computed using hospitals patients' data from 2019.



The spatially heterogeneous impact of COVID-19 could, in turn, worsen overall spatial inequality in the future. Preliminary studies suggest that the pandemic is likely to deepen economic inequality across individuals in Latin America and the Caribbean (Bottan et al., 2020; Busso et al., 2020). At the same time, some of the most affected regions include those that had high levels of inequality to begin with. This can be seen, for the case of Brazil, in the map in **Figure 21**. The map shows the spatial variation of COVID-19 cases and inequality—measured by the Gini coefficient of hourly wages. The spread of cases was particularly severe in the most populated cities, located in the South and Southeast regions. However, in other regions with relatively smaller urban areas in terms of population, there was a considerably large number of cases per capita. Northern cities with high inequality levels seem to have suffered a disproportionately higher incidence than comparable cities in other Brazilian regions. Although further research in this area is needed, these patterns suggest that COVID-19 could widen pre-pandemic spatial inequalities. Early evidence from the metropolitan area of Santiago, Chile, where pre-COVID-19 inequities have worsened since the beginning of the pandemic (Bilal et al., 2021a), also points in this direction.

Figure 21 ***Inequality and impact of COVID-19 across Brazilian cities***

Note: Authors' calculation based on data from Chauvin (2021b). Cumulative cases of COVID-19 are computed for 1,469 cities with a population of at least 20 thousand. COVID-19 data as of March 7, 2021.



8.3.3.5 Spatial health disparities within cities

Variation in health outcomes across sub-city units

Health disparities within cities can also be remarkably large. **Figure 22** illustrates this with data of 39 metropolitan sub-city units from São Paulo and 40 sub-units from Bogotá. In Bogotá, although there are no large differences in the incidence of some noncommunicable diseases like diabetes and kidney failure, for others there are considerable disparities. For example, the incidence of cardiovascular diseases in the most affected sub-city unit is double the prevalence of the least affected one. And this is also observed for other health-related behaviors such as smoking. In the case of São Paulo, gaps in the incidence of disease among patients are especially large for cardiovascular, diabetes, kidney diseases, and cancer with possible immunosuppression. In particular, in the sub-city unit with the highest incidence of cardiovascular illnesses, the difference is 35 percent higher than in the sub-city unit at the 25th percentile. For kidney failure and diabetes the difference between sub-city units at the 95 percent percentile can be more than four times the prevalence in the least affected one.

Figure 22 ***Differences in health outcomes across sub-city units in Bogotá and São Paulo***

Note: Bogotá's metropolitan area includes the 20 localities of Bogotá and 20 surrounding municipalities. Localities include 1922 neighborhoods. Localities and municipalities are weighted by population. Data come from the Multipropósito Survey (SDP, DANE, and Gobernación de Cundinamarca, 2017). São Paulo's metropolitan area includes São Paulo and 38 neighboring municipalities. With data from Chauvin, 2021b, prevalence of diseases per patient is computed using hospitals patients data. This figure does not include controls since its purpose is merely descriptive.



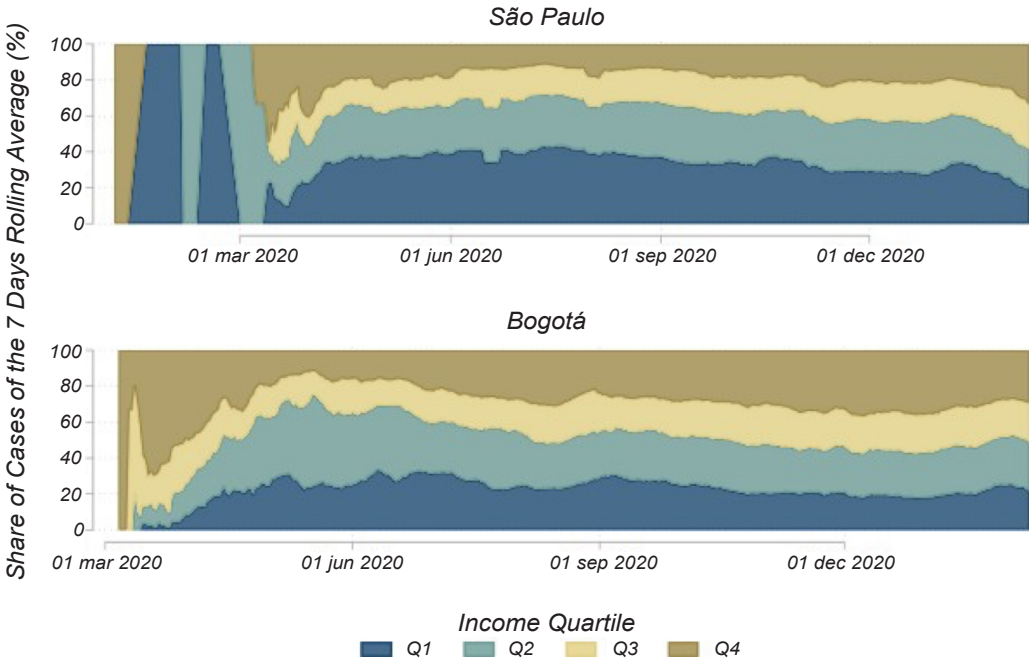
As in other regions of the world, in Latin America, within-city disparities in health outcomes are closely related with neighborhood characteristics. For example, across 50 neighborhoods of Bogotá, Parra et al. (2010) find that having safe parks and safety from traffic is positively associated with health-related quality of life (HRQOL) and self-rated health (SRH) among older adults. In turn, street noise had a negative effect on SRH. In São Paulo, Andrade et al. (2014) found a significant relationship between mental disorders and neighborhood social deprivation—an index that considers income, education, and family size. Exposure to crime was also correlated with anxiety and mood swings. Furthermore, these disorders more often affected vulnerable groups such as women and migrant men.

8.3.3.5.2 Neighborhoods and the spread of COVID-19

In Bogotá and São Paulo, the evolution of COVID-19 cases across income quartiles shows that the virus initially took root in relatively more affluent neighborhoods, where international travelers—who brought the disease to the country (Candido et al., 2020)—are more likely to reside. However, that changed rapidly over the following weeks. **Figure 23** illustrates this with data from metropolitan sub-city units. While the first infection occurred earlier in São Paulo than in Bogotá, in both cities the initial cases were concentrated in sub-city units belonging to the highest income quartile. Over time, however, the transmission of cases quickly affected other areas at different income levels, in particular, those from the lower-income quartiles. And in both cities, low-income neighborhoods (those in quartiles 1 and 2) bore most of the burden of the pandemic through the end of the first wave. Toward the end of 2020, new cases became more uniformly distributed among localities, particularly in Bogotá. Across sub-city units in São Paulo, the top 25 percent continued to have a relatively lower share of cases than the other quartiles for most of the observed period.

Figure 23 ***The spread of COVID-19 across neighborhoods in São Paulo and Bogotá***

Note: Authors' calculations using COVID-19 data of Secretaria Municipal da Saúde (2021) and SALUDATA (2021) for the 96 districts of São Paulo and the 20 Localities of Bogotá. Income data are taken, respectively, from the Secretaria Municipal da Saúde (2018) and the Multipropósito Survey (SDP, DANE, and Gobernación de Cundinamarca, 2017).



8.3.4 What explains spatial health inequalities in the region?

8.3.4.1 The role of the urban environment

8.3.4.1.1 Health and the local natural environment

A key determinant of health outcomes of the population of a geographic area—either urban or rural—is its natural environment. Temperatures, precipitation levels, and altitudes are closely tied to a city location, and they can affect the population's factors in multiple ways. Take for example a city's vulnerability to tropical diseases. Many cities in Latin America and the Caribbean are located in the tropics, where weather conditions expose their inhabitants to mosquito-borne infections, such as malaria, dengue, and zika. Among Colombian municipalities, areas with a higher rainfall variability had an average of four more cases of dengue per month—70 percent of the mean (De la Mata and Valencia-Amaya, 2014). Dengue has been associated with alterations in the immunological system (García et al., 2011), which could result in subsequent development of more severe diseases. Several studies find that in utero and postnatal exposure to such tropical diseases can indeed have long run effects. For example, adults who had malaria during childhood are more likely to interrupt their education earlier and to be illiterate (Barreca, 2010; Carrillo, 2020; Lucas, 2010). They also perform worse in cognitive tests (Venkataramani, 2012) and tend to have lower incomes (Bleakley, 2010).

Even relatively small atmospheric variations can have important health consequences. For example, evidence from the United States suggests that extreme temperatures can have negative effects on birth weight (Deschênes et al., 2009) and lead to higher mortality rates (Barreca, 2012; Deschenes and Moretti, 2009). And cities located at high altitudes, such as Mexico City, La Paz, or Quito, can be more vulnerable to particulates' pollution because motor engines are less efficient there than in cities at lower altitudes (Giraldo and Huertas, 2019).

Latin America and the Caribbean region is also vulnerable to major natural disasters such as earthquakes, hurricanes, and floods. Such events can impact local health outcomes in a variety of ways. Using data of disasters in Latin America over the last 100 years, Caruso (2017) finds that tropical cyclones increase the probability of being disabled. Earthquakes have had negative effects on child nutrition in Colombia (Bustelo et al., 2012), and similar results have been found in Nicaragua after Hurricane Mitch (Baez et al., 2010). The frequency and severity of these events are expected to increase over the next decades as a result of climate change (Pachauri and Meyer, 2014; UNDRR, 2019), and they will pose increasingly higher risks for cities in the region (Harlan and Ruddell, 2011; McMichael and Lindgren, 2011).

The ultimate impacts on health of risks related to the natural environment—such as tropical diseases, temperature fluctuations, or natural disasters—are not independent of human action. They are mediated by the preventive measures adopted by communities, including investments in adequate public infrastructure and preventive health services. But the financial burden that natural risks bring about can be dramatically higher in some localities than in others, and in many cases they can exceed the fiscal capacity of local and regional governments.

8.3.4.1.2 Health and urban built environment

In addition to their natural environment, health outcomes in urban areas are also affected by the built environment. Urban infrastructure, especially related to water, sanitation, and mobility, can have a substantive impact on individual and community health outcomes (WHO and UN-HABITAT, 2016).

Access to safe drinking water is tightly linked to the health outcomes of a city's population. There is ample evidence of this link in Latin America. During the 1990s, following a series of privatizations of water companies in Colombia, Granados and Sánchez (2014) document that, even though improvements in overall water access were limited, cities that received less of this service after the reform had smaller reductions in infant mortality. During the same period and following a similar privatization process in Argentina, child mortality decreased by about 8 percent, and in the poorest areas the impact was close to 26 percent (Galiani et al., 2005). Gamper-Rabindran et al. (2010) also show, this time using data from Brazil, that providing access to piped water reduces infant mortality, particularly in municipalities with the highest levels of under-one-year mortality rates.

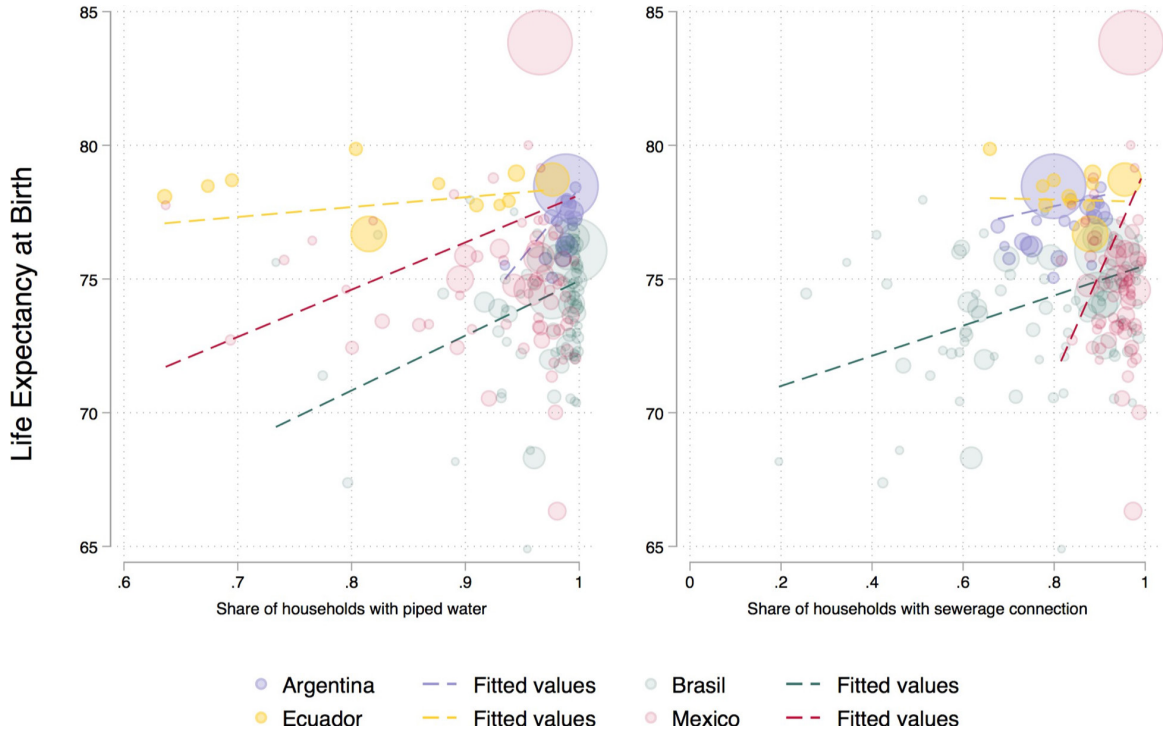
Across Latin American cities, access to safe drinking water is also linked to a higher life expectancy. This is illustrated in Figure 24 (left), which uses data from 177 cities with a population of at least 200,000, located in four different countries in the region. To different degrees, access to piped water is positively linked to life expectancy across cities of the same country. Studies from other regions of the world, especially in Africa and Asia, also find positive effects of water interventions on the prevalence of communicable diseases such as diarrhea and typhoid fever (Ashraf et al., 2017; Duflo et al., 2015; Kosec, 2014).

Access to infrastructure that allows for an adequate treatment and disposal of human excreta and sewage is another key component of a healthy urban environment. By 2025, it is estimated that municipal solid waste will increase by 70 percent globally to 2.2 billion tons/year, and despite the recent progress in sanitation coverage (Hoornweg and Bhada-Tata, 2012), many cities still lack proper management of such disposals. In Latin America, the cost-benefit ratio of water and sanitation improvements has been estimated to be around 5.0, suggesting large gains from a social perspective (Hutton, 2012). In cities without an adequate residuals management system, the population is more vulnerable to transmittable diseases, and they run the risk of developing long-term health conditions. For example, infant mortality has been associated with open defecation (Geruso and Spears, 2018), especially in cities with higher population density (Hathi et al., 2017).

Figure 24 (right) shows that, in most countries, cities where a larger share of the population has access to a sewerage connection tend to also have a higher life expectancy at birth. Other studies have also documented this relationship. For instance, Ortigoza et al. (2020) show that increased access to piped water and sewage significantly reduces infant mortality rates in a sample of 286 Latin American cities with populations above 100,000. Their estimations suggest that sanitation improvements can reduce infant mortality rates

Figure 24 ***Correlates of mortality rates across cities in four Latin American countries***

Note: Correlates are computed for 117 Latin America cities with populations greater than or equal to 200,000 (22 from Argentina, 81 from Brazil, 11 from Ecuador, and 63 from Mexico). Cities are weighted by population. The definition of city is comparable across countries. The estimation of life expectancy is done under standard life tables as defined by WHO (2014). Unlike in Bilal et al. (2021b), age-mortality rates are not discounted by mortality patterns or potential errors in the registration. Data from Chauvin (2021a).



between 15.5 and 6.2 percent. These patterns do not seem to be driven by sample composition: in the context of 363 cities of the continent, Bilal et al. (2021b) also find that a higher life expectancy is related to a higher access to water and sanitation, among other socioeconomic factors. Galdo and Briceño (2005) find that in Quito, Ecuador, improvements in sewage coverage reduced child mortality.

The urban built environment can also facilitate or constrain the adoption of healthier lifestyles among individuals (Smith et al., 2012). For example, access to public transit can influence the amount of physical activity that individuals perform. In the United Kingdom, switching from a passive transport mode such as cars to active commuting (i.e., walking or cycling) or to public transportation is associated with an average reduction of 0.3 kg/m^2 in the body mass index (BMI). In turn, switching to cars leads to an equivalent increase in BMI (Flint et al., 2016). In the United States, the decline in density and the emergence of suburban areas brought about a higher reliance on vehicles and subsequent increases in obesity (Zhao and Kaestner, 2010).

8.3.4.1.3 Transport infrastructure and the spread of transmissible diseases

Another type of infrastructure with strong links to local health is that related to transport, especially one that improves the connectivity between cities. This can have many and sizeable positive effects on the economy and on health (ITF, 2018; Leduc and Wilson, 2012), which are discussed in section 8.3.4. But it also makes cities more interdependent and facilitates the diffusion of transmissible diseases across space. Historically, cities that were more connected were the first to be affected by pandemics such as the Black Death in Europe (Jedwab et al., 2020). Outbreaks that took place in more recent periods, such as the Spanish Flu (1918), H1N1 (2009), Ebola (2014–16), spread more rapidly around the world, thanks in no small part to the modern global transport infrastructure (Abhishek et al., 2014; Basco et al., 2021).

Transport connectivity also played a key role in the diffusion of COVID-19. This was the case in Brazil, where the virus spread after the arrival of infected international travelers coming from western Europe. Southeast cities with large populations and strong international connections, such as São Paulo, Rio de Janeiro, and Belo Horizonte, were precisely the first to experience infections (Candido et al., 2020).

Table 3 presents regression estimates of the correlations between different types of connectivity and the initial arrival of COVID-19 across Brazilian cities. In this analysis, the dependent variable is the number of days that the city remained with zero cases since January 1, 2020. The first two columns show the correlations of this outcome with measures of international air connectivity, including the distance to the closest international airport and the number of flights that arrived at that airport in 2019, weighted by distance. Column (1) estimates the partial effect of these international connectivity indicators, without including other controls. It shows that urban areas that were more distant from international airports or had fewer international travelers remained COVID-free for a longer time. Column (2) adds population variables to the basic model. Once population controls are included, the size and statistical significance of international connectivity measures drop significantly, which highlights the fact that more populated cities tend to also be more internationally connected. Moreover, the negative and significant association with population variables indicates that more populated and denser cities were more likely to experience cases early in the pandemic.

Once COVID-19 entered Brazil, national connectivity also played a role in its diffusion. Columns (3) to (6) from Table 3 show the impact of various measures of domestic air and land connectivity on the time elapsed until a city reported its first COVID-19 case. Similar to the results of international connectivity, cities with a higher number of domestic flights—columns (3) and (4)—reported the first case of COVID earlier than others, and this correlation remains even after including population controls. The last two columns focus on measures of land connectivity computed by the Brazilian Institute of Geography and Statistics (IBGE, 2017). Connectivity degree is defined as the number of cities that have a direct connection to the focal city, while the proximity index is calculated as the inverse of the average distance to the rest of the cities in the national road network. These results are also consistent with the estimates of air connectivity.

Table 3 Delays in COVID-19 first confirmed case and city connectivity in Brazil

Logarithm of days between January 1, 2020, and the first confirmed case in the city

	(1)	(2)	(3)	(4)	(5)	(6)
International Air Connectivity						
Distance to international airport	0.055***	−0.000				
	(0.010)	(0.006)				
International flights	−0.004**	−0.002				
	(0.002)	(0.002)				
Domestic Air Connectivity						
Distance to domestic airport			0.063***	0.009		
			(0.006)	(0.006)		
Domestic flights			−0.006***	−0.002*		
			(0.002)	(0.001)		
Land Connectivity						
Connectivity degree					−0.062***	−0.007
					(0.009)	(0.005)
Proximity Index					−0.196**	−0.025
					(0.086)	(0.053)
Population Controls						
Log of population		−0.106***		−0.102***		−0.099***
		(0.008)		(0.009)		(0.008)
Log of population density		−0.036***		−0.034***		−0.037***
		(0.005)		(0.004)		(0.004)
Observations	3,759	3,759	3,730	3,730	3,732	3,729
R^2	0.177	0.490	0.246	0.493	0.300	0.491

Source: Prepared by the author based on SINADEF, Secretaría de Salud del Gobierno de México, Estadisticas Vitales (DANE), Instituto Nacional de Estadistica (INE), Instituto Nacional de Estadística y Censos de Ecuador (INE), National Censuses of Colombia, Chile, Peru, Ecuador, El Salvador and Mexico. Last update: Dec 20th, 2021.

Notes: Estimations are restricted to cities with populations greater than or equal to 5,000. Air connectivity is defined as the distance to the closest international or domestic airport (in km), and the number of international or domestic flights weighted by distance. These variables are included as the natural logarithm. All models are estimated by OLS and include state fixed effects. Robust standard errors are in parentheses and clustered at the state level. Data comes from Chauvin (2021b).

*** $p < .01$, ** $p < .05$, * $p < .1$

8.3.4.2 Spatial disparities in access to health care

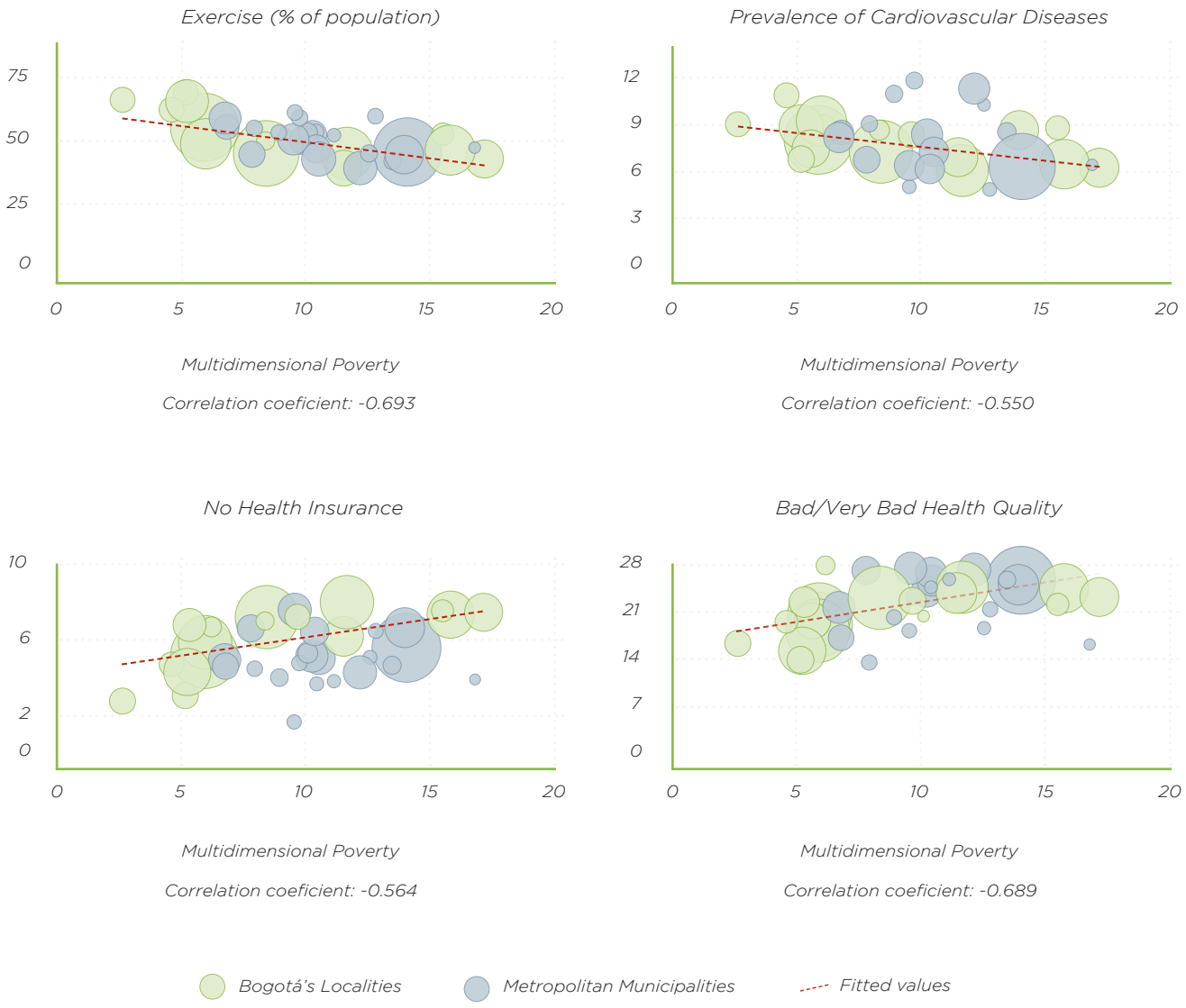
Compared to OECD countries, Latin America is still behind in terms of access to medical resources. For example, the region has on average two doctors per 1,000 population, in contrast with the 3.5 OECD mean. Additionally, out-of-pocket health expenditure is considerably higher (34 percent) than in rich countries (21 percent), reflecting insufficient health insurance coverage (OECD/The World Bank, 2020).

Within Latin American countries, access to health care differs significantly across cities. In a survey conducted by the Pan-American Health Organization, the percentage of respondents who reported getting an appointment within the same week was 59 percent in São Paulo, 82 percent in Montevideo, and around 67 percent in Santiago and Buenos Aires. There are also spatial variations in the lack of health insurance. In Buenos Aires, lack of health insurance coverage reaches 17 percent, in Santiago it is 11 percent, and in São Paulo and Montevideo it is about 2 percent (Balsa et al., 2011).

Sizable gaps in access to health care exist also across neighborhoods of the same city. Even in high-income countries like the United States, where connectivity infrastructure is better developed, mortality increases when there are fewer doctors within 25 miles of the patient (Li, 2014), or when the closest hospital is farther away (Buchmueller et al., 2006). The connection between physical access to health care centers and outcomes has also been documented in Latin American countries (Weis et al., 2020). For example, across Peruvian cities, increases in travel time to health care facilities have been associated with higher pneumonia fatality rates in under-five children (Carrasco-Escobar et al., 2020). Access to health insurance can also vary significantly within cities. This is illustrated in **Figure 25** using data of sub-metropolitan units (localities) from Bogotá, Colombia. The population of higher-poverty localities tend to have, at the same time, worse health outcomes and a higher likelihood of lacking health insurance.

Figure 25 ***Poverty and health across Bogotá's metropolitan area***

Note: Bogotá's metropolitan area includes the 20 localities of Bogotá and 20 surrounding municipalities. Localities gather include neighborhoods. Localities and municipalities are weighted by population. Bad health quality is defined as the percentage of individuals that considered their health insurance service as "bad" or "very bad." Exercise is measured as the proportion that reported doing physical activity at least once per week, during the last month. Data come from the Multipropósito Survey (SDP, DANE, and Gobernación de Cundinamarca, 2017) and the 2018 National Census (DANE, 2018).



8.3.4.3 Negative urban externalities

8.3.4.3.1 Congestion, pollution, and health

An important characteristic of urban areas that is also tied to health outcomes is population density. Density brings about many economic benefits, but at the same time creates conditions for individuals to negatively affect other people's welfare. Two such externalities are traffic congestion and air pollution. The decision of an individual to drive instead of taking public transit, for example, does not only contribute to traffic congestion and deficient road safety, but it also leads to the generation of polluting emissions and noise. Drivers typically do not consider these external effects when they decide to use their vehicles.

Urban air pollution is a growing concern in many Latin American cities. While the WHO target for average of particulate matter (PM_{10}) is 35 Qg/$m^{3,}$ multiple large cities in the region have recorded particulate levels well above that target, including Santiago in Chile (64), Mexico city (56), and Bogotá (52) (Riojas-Rodríguez et al., 2016). Several studies have shown a clear negative impact of air pollution on health outcomes. In Mexico City, using a 10-year panel of 48 municipalities, Arceo et al. (2016) found that increasing 1 Qg/m3 in 24-hour particulate matter increases weekly infant deaths per 100,000 births by 0.23. Further evidence from US counties also shows that air pollution is related with higher child mortality rates (Chay and Greenstone, 2003).

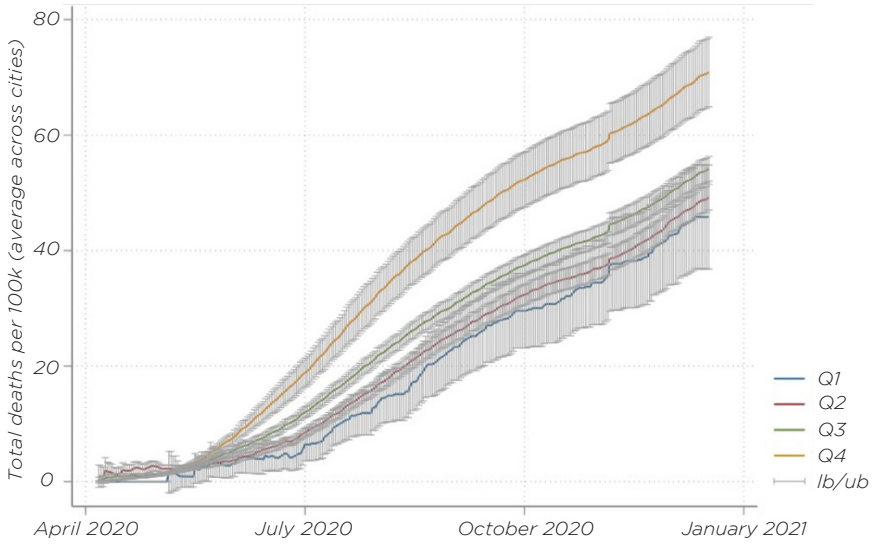
8.3.4.3.2 Urban characteristics and the spread of COVID-19

Another well-known externality associated with high-density environments is the spread of infectious diseases. The likelihood of contagion of such diseases typically grows with the number of human interactions, and interactions are, in turn, an exponential function of total population (Schläpfer et al., 2014).

Such patterns have been observed during past pandemics and recently confirmed by the spread of COVID-19. As with other infectious diseases, the size of a city's population was an important predictor of the local impact of the pandemic. **Figure 26** illustrates this, showing the evolution of weekly deaths per 100 thousand people in Brazilian cities between April 2020 and January 2021, separately for each city population quartile. As discussed above, the virus arrived earlier in more populated cities, which also tend to be more internationally connected. This meant that, from the very beginning, large cities were more affected by COVID-19, both in absolute and in per-capita terms. Throughout the first year of the pandemic, cities in higher population quartiles systematically reported more deaths per capita than cities in lower population quartiles. And the initial gap, rather than closing as smaller cities started to also report infections and deaths, grew over time.

Figure 26 **Impact of COVID-19 in Brazil over time by city size quartiles**

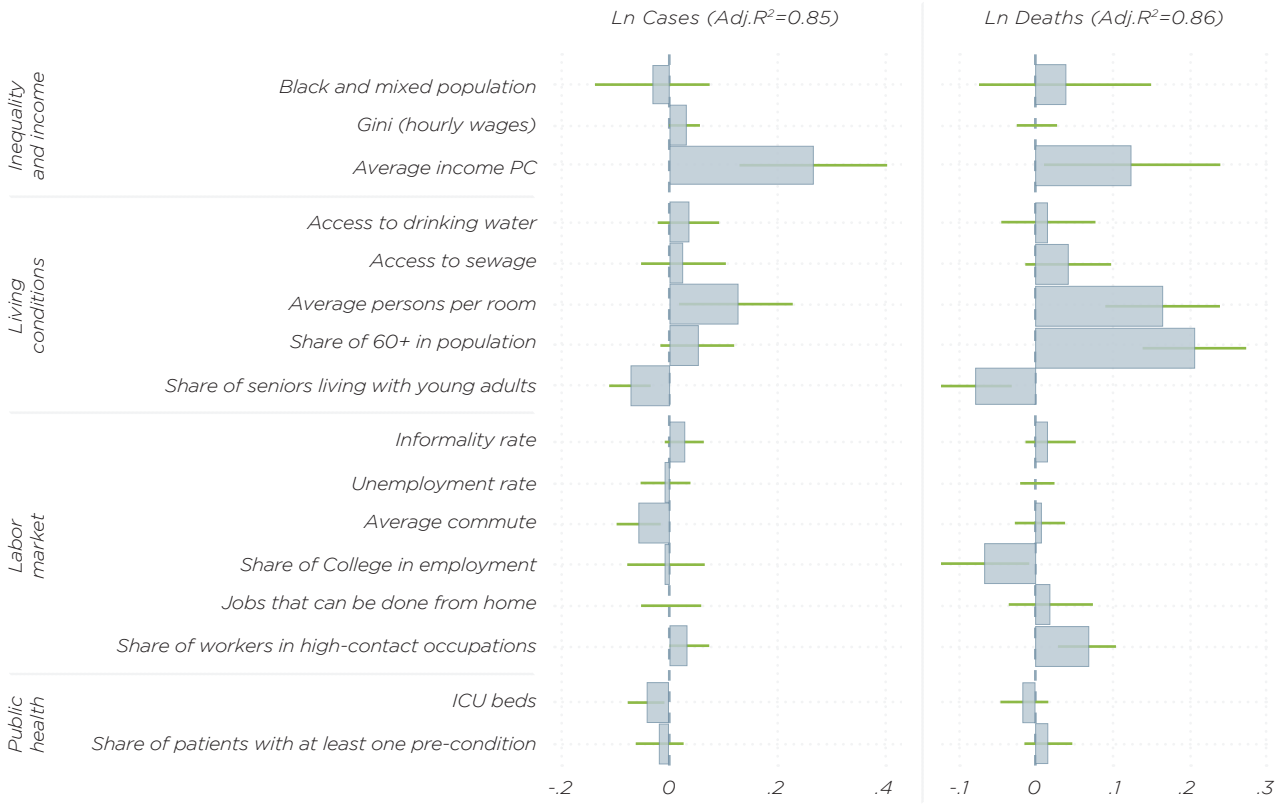
Note: Each line represents the average evolution of deaths per capita across cities at each population quartile. Vertical lines correspond to the 95 percent confidence intervals. The impact of COVID-19 is computed for 3,762 cities using data from Chauvin (2021b).



But population and density are not the only city characteristics that facilitated the spread of COVID-19. To shed light on what other factors explain why the pandemic affected some places more than others, **Figure 27** reports regression estimates of the effects of four sets of variables on city-level cumulative COVID-19 cases and deaths per capita by January 2021, using data of 2,439 Brazilian cities from Chauvin (2021b).[4] Four sets of socioeconomic characteristics are considered: (1) inequality and income, (2) living conditions, and (3) labor market, and (4) public health.

Figure 27 Correlates of the local impact of COVID-19 in Brazil

Note: Each bar represents the point estimates for each regressor. Respectively, the dependent variables are the natural logarithm of the total number of confirmed cases and confirmed deaths from COVID-19 in each city. Data are updated until January 20, 2021. Estimations also control for population (which allows for interpreting the results in per capita terms), population density, urbanization, the number of international passengers and the number of national passengers arriving at the closest airport in 2019 (as indicators of exposure to air travelers), average maximum yearly temperature, average yearly precipitation, broadband connections density, the share of domestic workers in local employment, the percentage of the population living in slums, and distance to São Paulo. It also includes other baseline public health variables, including the number of doctors and the number of ventilators. All variables are standardized. Regressions are estimated by OLS and include state fixed effects. Lines represent 95 percent confidence intervals. Robust standard errors are clustered at the state level. Estimations are restricted to 2,439 cities with population greater than or equal to 10,000. Data come from Chauvin (2021b).



4. Chauvin (2021b) shows that the effects of city characteristics on local cases and deaths can fluctuate over time. However, for most variables the direction of the effect is consistent throughout the first year of the pandemic.

These estimates show that, contrary to what has been observed in the United States (Desmet and Wacziarg, 2021), cities with higher average income per capita in Brazil were relatively ***more*** affected by COVID-19, in terms of both cases and deaths, after controlling for all other covariates. Chauvin (2021b) shows that higher-income cities also experienced higher mobility, as a smaller share of the population stayed at home for most of this period. In contrast with cities in high-income countries, in Brazilian cities a large part of the population is unable to perform their work from home and lacks the savings to support prolonged periods of economic inactivity. Cities with higher income are also characterized by a relatively higher demand for goods and services, which translates in a higher availability of jobs and creates the incentives for increased mobility.[5] This is also consistent with the fact that, as shown in Figure 27, cities with larger shares of workers in high-contact occupations also experienced more COVID-19 deaths per capita. Note that, as previously discussed, this is not necessarily the case when we consider the variation within metropolitan areas. Richer neighborhoods within São Paulo, for example, were relatively less affected by COVID-19 on a per-capita basis. This suggests that low-income populations living in more affluent cities were particularly vulnerable to the pandemic.

Another important result related to the urban environment that comes out of this analysis is that housing conditions also shaped the local impact of COVID-19. In particular, a factor that appears to have played an important role after accounting for all other covariates is residential overcrowding (as measured by the number of persons per room living in the average household). Cities with more residential overcrowding experienced more infections and deaths than other cities.

Additional results highlight that local vulnerability to the pandemic was also related to the composition of the local population, and in particular with the share that was at increased risk. For example, controlling for all other variables, cities with older populations—which, before vaccinations, were at significantly higher risk of developing severe complications from the disease (Davies et al., 2020; Levin et al., 2020)—experienced more deaths per capita. The presence of higher-vulnerability people may have also created incentives for precautionary behavior in the rest of the population, as suggested by the fact that fewer deaths per capita were observed in cities with large shares of households where older and younger generations cohabitate.

8.3.5 Health inequalities and local public policy

Even though addressing spatial health inequalities requires action at different geographic scales, there are multiple policy instruments that local governments can deploy and that can lead to meaningful improvements. Among the most effective, as discussed above, is the provision of basic public services such as water and sewerage. This is also a development outcome in itself, and one of the United Nations' Sustainable Development Goals. But there are other interventions that can also contribute to reducing spatial inequalities, directly or indirectly.

One example is public transit, which may have had an adverse effect in the propagation of infections during the COVID-19 pandemic, but has been shown to have positive health effects in normal times through decreased pollution externalities. A study covering 58 countries found that subway openings between 2001 and 2016 reduced particulates by 4 percent in highly polluted cities. The study also estimates that, for a typical city in this sample (population of 5.3 million and a 2 percent birth rate), a new subway system can avoid around 22–34 infant deaths each year (Gendron-Carrier et al., 2021). Ortigoza et al. (2020) also find, looking at Latin American cities, that an improvement of one standard deviation in mass transit availability can decrease infant mortality rates by 6.6 percent. Another benefit of public transit projects is that they can create incentives for increased physical activity even in contexts where large shares of the population walk or rely on informal transportation. In Bogotá, Lemoine et al. (2016) found that the implementation of the rapid bus system—TransMilenio—was related to an increase of physical activity of around 22 minutes/day across income groups.

Other, less costly local policies can also promote the adoption of healthier lifestyles in the population, particularly among the more economically disadvantaged. One example is the construction of recreational infrastructure. In Brazil, for instance, neighborhoods with parks or other places dedicated to physical activity tend to have a lower proportion of people with overweight (Velásquez-Meléndez et al., 2013). Another example is the implementation of programs promoting the use of bicycles for commuting and recreation.

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5 In principle, this correlation could also reflect the fact that higher-income cities may also have had more testing capabilities, particularly at the beginning of the pandemic. However, this does not appear to be the main driver of the observed result. The effect of city income did not significantly shrink over time—as access to testing became more available throughout the national territory—and its magnitude remains very similar when it is calculated comparing only cities belonging to the same state, even though testing regimes were significantly more homogeneous within than across states (Chauvin, 2021b).

The *Muevete en Bici* program implemented in Mexico City increased vigorous physical activity by 71 minutes/week (Medina et al., 2019), and estimates suggest that the costs-benefits of the construction of bike lanes are positive in various cities, including Bogotá (3.23–4.26), Medellín (1.8), and Guadalajara (1.02–1.23) (Montes et al., 2012).

Air and noise pollution have also been addressed through circulation restrictions, but evidence of their effectiveness is more mixed. For instance, the *Hoy No Circula* program from Mexico City, which bans the use of cars on specific days based on the plate's last digit, created incentives for richer households to purchase additional vehicles, resulting in an increase of the total volume of circulating cars, and no reductions in air pollution levels (Davis, 2008). Similar effects were found in Colombia, where the *Pico y Placa* program has not improved air quality but has led to increases in gasoline consumption and vehicle ownership in Bogotá (Bonilla, 2019), and to changes in commuting departure times in Medellín (Ramos et al., 2017). An exception is Quito, where a similar circulation restriction program led to improvements in the concentration of carbon monoxide during rush hours (Carrillo et al., 2016). According to more extensive review by Blackman et al. (2018) and IADB (2020), a robust enforcement combined with low cost-effective public transportation becomes critical to successfully implement these programs.

Another policy frequently pursued by local governments that has been shown to have positive effects on health outcomes is slum upgrading programs. For example, in Mexico and El Salvador, the TECHO program, which helped improve the quality of construction in informal dwellings, reduced by 27 percent the under-five incidence of diarrhea, which is similar to the impact of water and sewage connections (Galiani et al., 2017).

Policies seeking to improve local health conditions, however, under some circumstances can also lead to undesired consequences. Bancalari (2020) provides a good illustration of this by studying 6,000 sewage projects implemented in Perú between 2005 and 2015.

Completed projects reduced infant mortality by 33 percent. Nevertheless, unfinished projects were associated with a 5 percent increase in infant mortality, as they exposed communities to water cuts and excavations-related accidents. This highlights that the effect of local interventions depends not only on choosing a technically adequate solution but also on how this solution is implemented.

8.3.6 Conclusion

In Latin America and the Caribbean, there are significant spatial disparities in health outcomes across and within cities. This article has provided an overview of the existing evidence regarding the magnitude of these disparities, what drives them, and what policymakers can do to address them.

A first conclusion is that spatial inequality in health outcomes is large in Latin America. The incidence of non-transmittable diseases, such as hypertension and obesity, can be more than three times higher in some cities in the region than in others, and the gap in life expectancy can reach more than eight years. Part of these figures reflect national-level differences, but for some diseases, more than half of the variation is driven by within-country differences. These large within-country spatial differences in health outcomes were also observed during the COVID-19 pandemic. In Brazil, for example, a city in the top quartile of confirmed COVID-19 cases per capita, reported about 3.5 times as many confirmed cases as in the cities in the bottom quartile during the first year of the pandemic.

The data and literature reviewed in this article also suggest that part of these geographic inequalities are explained by the fact that socioeconomically vulnerable individuals—who also tend to face more health challenges—tend to spatially cluster. However, the analysis also points to multiple local characteristics that, in turn, shape the ultimate health outcomes of each locality. These include aspects of the natural environment, such as the presence of tropical diseases, differences in temperature and altitude, and vulnerability to natural disasters, as well as aspects of the built environment that can be shaped through policy.

Local policymakers can play a key role in addressing spatial health inequalities. In addition to the construction of water and sanitation infrastructure, for example, they can promote the reduction of urban pollution and congestion externalities by investing in public transit systems and promoting physical activity among their population. Programs primarily motivated by other policy objectives, such as slums upgrading programs, can also have a significant impact on local health outcomes. Spatial inequality is an issue in which many opportunities exist to make a meaningful difference through deliberate and evidence-based local action.

KEY FINDINGS

Life expectancy, the burden of non-communicable diseases, and, more recently, the impact of COVID-19 all exhibit large geographic disparities both across and within cities.

Part of these geographic inequalities are explained by the fact that socioeconomically vulnerable individuals—who have, on average, poorer health—tend to spatially cluster.

However, characteristics of the local environment—including access to health services, infrastructure that favors healthy lifestyles, water and sanitation, and atmospheric conditions—can improve or further deteriorate health outcomes.

Existing policy tools available to local and national governments can be effective in reducing health inequalities by creating healthier built environments and reducing negative externalities, particularly in places where socioeconomically vulnerable communities reside.



Bogotá, Colombia

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